Type MS · Describe a trend or compare data sets and conclude

Answer 1 of 27 · Question 4 part a.iv · 2 marks

2024 JUN · Paper 01 (regional R) · find part a.iv in this mark scheme extract

| Question number | | | | |
|---|---|---|---|---|
| 4 | (a) (i) | • axes labels and units and right way around;<br>• linear scale and at least half grid;<br>• correctly plotted points to half square tolerance; | | (3) |
| | (ii) | • curve of best fit; | do not accept straight line going through all points | (1) |
| | (iii) | 6 | | (1) |
| | (iv) | • time taken to digest starch decreases until pH 6;<br>• time taken to digest starch increases (from/after PH6) to pH10; | Allow 1 mark for decrease and then increase in time as pH increases | (2) |
| | (v) | • reference to pH2 not being optimum/best pH for **enzyme** (activity)/pH6 being (closer to) optimum pH for enzyme;<br>• enzymes denatured / active site changes shape/deformed (at pH2);<br>• less/no binding to substrate/ substrate cannot fit into active site/ fewer enzyme -substrate complex; | | (3) |
| | (vi) | small intestine/ mouth/duodenum; | | (1) |
| | (b) | • add iodine solution;<br>• blue-black/blue/black colour indicates the presence of starch; | | (2) |

**Total for Question 4 = 13 marks**

## Type MS · Describe a trend or compare data sets and conclude

### Answer 2 of 27 · Question 3 part b.iv · 1 mark

2019 JUN · Paper 01 · find part b.iv in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Describe a trend or compare data sets and conclude

Answer 3 of 27 · Question 6 part a.i · 1 mark

2019 JUN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

<table>
<tr><td>6 (a) (i)</td><td>Increase/goes up;</td><td>allow equivalent</td><td>1</td></tr>
<tr><td>(ii)</td><td>Information required about other factors, such as diet/exercise/alcohol consumption/age/gender/exposure to pollution/(genetic) history/smokers or non smokers;;</td><td>Allow one mark for reference to lifestyles only</td><td>2</td></tr>
<tr><td>(b)</td><td>Less mucus removed by cilia/more mucus/build up of mucus in lungs/blocks airways/bacteria not removed;<br>Increased risk of (lung/respiratory) infection/coughing;</td><td></td><td>1<br>1</td></tr>
<tr><td>(c) (i)</td><td><table><tr><td></td><td>Surface area</td><td>Volume</td><td>SA:V ratio</td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td>54;</td><td>27;</td><td>2:1;</td></tr></table></td><td>Allow for SA:V ratio 6:3 ECF</td><td>3</td></tr>
<tr><td>(ii)</td><td>Smaller surface area to volume ratio (in lungs with emphysema);<br>Reduced/less efficient gas exchange/diffusion of oxygen/carbon dioxide;<br>Less oxygenated blood/less oxygen transferred (to cells);<br>Less (aerobic) respiration;</td><td></td><td>Max 3</td></tr>
</table>

## Type MS · Describe a trend or compare data sets and conclude

### Answer 3 of 27 · Question 6 part a.i · 1 mark

2019 JUN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Describe a trend or compare data sets and conclude

### Answer 4 of 27 · Question 3 part a.ii · 2 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

| Number | | | |
|---|---|---|---|
| 3 (a) (i) | • correct labelling of axes (1)<br>• suitable scale (1)<br>• X and Y axis correct way round (1)<br>• plotting (2) | minus 1 for each incorrect plot | 5 |
| (ii) | Any two from<br>• few at lower end/150-151/top end/160 (1)<br>• majority in middle/154-157 (1)<br>• increases to 154-155 then decreases (1) | | 2 |
| (b) | Any three from<br>• nutrition (1)<br>• sex (1)<br>• age (1)<br>• genes/genetically based (1) | | 3 |

Total 10 marks

## Type MS · Describe a trend or compare data sets and conclude

### Answer 4 of 27 · Question 3 part a.ii · 2 marks

2020 NOV · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Describe a trend or compare data sets and conclude

Answer 5 of 27 · Question 3 part c.i · 2 marks

2022 JAN · Paper 02 · extract 1 of 2 · find part c.i in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 3 | (a) | any two from<br>• protection/named example/barrier against pathogens;<br>• sensory reception/named example;<br>• excretion | | 2 |
| | (b) | • looks red;<br>• vasodilation/more blood in skin <u>capillaries</u>/blood flows closer to skins surface;<br>• allows more heat loss/heat energy released;<br>• to lower body temperature; | | 4 |
| | (c) (i) | • as temperature increases, rate of sweating increases;<br>• up until 34°C/ between 32 °C - 35.9 °C, little/no change;<br>• increases, rapidly/significantly from 35.9 °C - 38 °C; | Allow any temperature between 34°C and 35.9 °C for 35.9 °C e.g. between 32°C and 34 °C | 2 |
| | (ii) | • absorb sweat with cotton wool/absorbent paper;<br>• fixed area of skin/under arm pits;<br>• reference to measuring mass of absorbent material before and after;<br>or<br>• measure time;<br>• for cobalt chloride paper to change colour; | | 2 |
| | (d) | • enzymes affected/to maintain enzyme activity/ref to body temperature being optimum for enzymes;<br>• too high a temperature denatured/prevent denaturation;<br>• too low, inactive/less active/prevent enzymes becoming inactive; | | 3 |

Total 13 marks

## Type MS · Describe a trend or compare data sets and conclude

### Answer 5 of 27 · Question 3 part c.i · 2 marks

2022 JAN · Paper 02 · extract 2 of 2 · find part c.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Describe a trend or compare data sets and conclude

Answer 6 of 27 · Question 3 part a.iii · 3 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

| Question number | | | | | |
|---|---|---|---|---|---|
| 3 | (a) | (i) | • time in minutes;<br>• 0-10, 11-20, 21-30, 31-40,41-50,51-60; | | 2 |
| | | (ii) | • 75 + 72 + 68; plus/minus 2 for each one<br>• 215 $cm^3$; | ECF | 2 |
| | | (iii) | • rapid reaction/a lot/large amount of gas produced, at start;<br>• rate slows/less gas (for next 10 minutes);<br>• drops a lot/little gas produced (after 40 minutes); | | 3 |
| | | (iv) | any three from<br>• at start plenty of substrate/many E-S complexes formed;<br>• substrate used up/less substrate available;<br>• reaction slows/less gas/product, produced/less enzyme action;<br>• most substrate used at end so little gas produced; | | 3 |
| | (b) | | • less gas produced/reaction lower/less;<br>• because active sites blocked; | | 2 |

Total 12

## Type MS · Describe a trend or compare data sets and conclude

### Answer 6 of 27 · Question 3 part a.iii · 3 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Describe a trend or compare data sets and conclude

Answer 7 of 27 · Question 4 part b.i · 2 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 4 | (a) | any two from<br>• hearing/sound;<br>• touch;<br>• taste;<br>• smell; | | 2 |
| | (b) (i) | any two from<br>• as thickness decreases, focal length increases;<br>• interval between focal lengths increases as thickness decreases;<br>• (constant when two variables multiplied hence) inversely proportional; | | 2 |
| | (ii) | diameter decreases; | | 1 |
| | (iii) | any four from<br>• rays from near object diverging;<br>• ciliary muscle contracts;<br>• suspensory ligaments slacken/relax;<br>• so lens becomes thicker/more convex;<br>• more powerful lens/light refracted more/increased refraction;<br>• (allows it to focus because) focal length becomes shorter/decrease; | | 4 |
| | (c) (i) | between 575 and 580 nm; | | 1 |
| | (ii) | • 400 - 530;<br>• 130 nm; | ECF | 2 |
| | (iii) | • greater range of wavelengths can be seen/absorb light of different wavelengths;<br>• gives better colour perception/vision/differentiation/can see different colours; | | 2 |

Total 14

## Type MS · Describe a trend or compare data sets and conclude

### Answer 7 of 27 · Question 4 part b.i · 2 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Describe a trend or compare data sets and conclude

Answer 8 of 27 · Question 2 part b.i · 3 marks

2023 JUN · Paper 02 (regional R) · find part b.i in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 2 (a) | (i) | in the following order:<br>• proteases;<br>• protein Q;<br>• amino acids; | Reject protein P | 3 |
| | (ii) | stomach/ small intestine/ pancreas/ duodenum/ ileum; | | 1 |
| | (iii) | DNA / mRNA / messenger RNA; | | 1 |
| (b) | (i) | • as pH increases activity increases;<br>• to pH2/peaks at pH2/to optimum;<br>• decreases from pH2; | Allow increases and then decreases for one mark | 3 exp |
| | (ii) | Any two from:<br>• enzyme is denatured (at pH2);<br>• change in shape of active sites;<br>• no enzyme-substrate complexes formed/no binding with substrate; | | 2 exp |
| | (iii) | • repeat;<br>• calculate a mean/average/remove anomalous result | | 2 grad |
| | (iv) | Any one from:<br>• temperature;<br>• substrate concentration/volume;<br>• enzyme concentration/volume; | | 1 grad |

**Total for Question 2 = 13 Marks**

Type MS · Describe a trend or compare data sets and conclude

Answer 9 of 27 · Question 6 part b.i · 1 mark

2024 JUN · Paper 02 (regional R) · find part b.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 6 (a) | • reduced blood flow to the heart/cardiac muscle/coronary arteries;<br>• less oxygen to heart/cardiac muscle/tissue;<br>• less (aerobic) respiration/less energy for contraction;<br>• (cardiac) muscle/cells die;<br>• (leading to) heart attack/heart disease; | | (4) |
| (b) (i) | As the use of statins increases the level of blood cholesterol decreases; | allow inversely proportional | (1) |
| (ii) | 1988 = 31;<br>2005 = 17;<br>1 : 1.8/1 : 0.54/1:0.548/1:0.55; | ecf for incorrect graph readings.<br>if use 18:17 ratio is 1:1.0588 (accept 1:1.059, 1.06, 1.1)<br>full marks for correct final answer | (3) |
| (iii) | • select sample of people;<br>• record the number of people taking statins/not taking statins;<br>• take blood samples;<br>• measure the concentration/level of cholesterol (in blood);<br>• repeat every year/between 1988-2008; | | (5) |
| (iv) | any two from the following:<br>• gender/age not taken into account;<br>• sample size maybe too small;<br>• reference to a named lifestyle not taken into account e.g. diet, level of activity/smoking;<br>• genetic predisposition/diseases not taken into account; | | (2) |

**Total for Question 6 = 15 marks**

Type MS · Describe a trend or compare data sets and conclude

Answer 10 of 27 · Question 1 part c.i · 2 marks

2024 JUN · Paper 02 · find part c.i in this mark scheme extract

| Number | | | |
|---|---|---|---|
| 1 (a) | <table><tr><th>Disease</th><th>Type of pathogen</th></tr><tr><td>cholera</td><td>Bacterium/bacteria;</td></tr><tr><td>AIDS</td><td>virus;</td></tr><tr><td>athlete's foot</td><td>Fungus/fungi;</td></tr></table> | Reject more than one line from any disease | (3) |
| (b) | In the following order:<br><br>bacterium;<br>sexual intercourse;<br>barrier;<br>antibiotics; | | (4) |
| (c) (i) | • overall decrease;<br>• fluctuates<br>• steep decreases from 1990; | | (2) |
| (ii) | 17 - 9;<br>8; | Allow tolerance of +-0.5<br><br>ecf for mp2 | (2) |
| (d) | C antigens,white blood cells produced,antibodies destroy pathogens;<br><br>The following are incorrect because -<br>A vaccinations do not contain antibodies<br>B platelets are not produced following a vaccination<br>D vaccinations do not produce platelets | | (1) |

**Total for Question 1 = 12 marks**

## Type MS · Describe a trend or compare data sets and conclude

### Answer 11 of 27 · Question 2 part a.iii · 1 mark

2024 NOV · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

| Number | | | | | |
|---|---|---|---|---|---|
| 2 | (a) | (i) | • axes labels;<br>• plots;<br>• scale; | axes correct way round<br>use aty least half of the graph paper for the plotted results<br>ignore extrapolation of line | 3 |
| | | (ii) | line of best fit; | | 1 |
| | | (iii) | As time increases pH decreases; | ECF from an incorrectly drawn graph | 1 |
| | (b) | | Any three from:<br>• lipase;<br>• digests/breaks down fat (in milk);<br>• to fatty acids;<br>• which lowers pH/increases acidity; | | 3 |
| | (c) | | time; | allow mins/minutes | 1 |
| | (d) | | Any two from:<br>• temperature;<br>• volume/concentration of enzyme/substrate (solution)/volume of milk;<br>• type of milk; | ignore amount | 2 |
| | (e) | | repeat the experiment | | 1 |

**Total Marks 12**

## Type MS · Describe a trend or compare data sets and conclude

### Answer 11 of 27 · Question 2 part a.iii · 1 mark

2024 NOV · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Describe a trend or compare data sets and conclude

Answer 12 of 27 · Question 4 part a.i · 2 marks

2025 JUN · Paper 02 (regional R) · find part a.i in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 4 | (a) | (i) | • decreases;<br>• from 82% to 72%/by 10% | | 2 |
| | | (ii) | 79:21;<br>3.76:1; | accept 3.7612:1<br>accept 3.762:1<br>accept 3.8/4:1 | 2 |
| | | (iii) | any two from<br>• passive smoking/ex- smokers already have damaged lungs;<br>• (air) pollution;<br>• genetics/heredity; | | 2 |
| | | (iv) | tar; | | 1 |
| | (b) | (i) | bronchiole; | | 1 |
| | | (ii) | alveolus; | | 1 |
| | | (iii) | any four from<br>• damaged alveoli/alveoli walls broken;<br>• less surface area (for gas exchange);<br>• difficulty breathing/shortness of breath;<br>• less oxygen passing into the blood/ less carbon dioxide removed from blood/less gaseous exchange;<br>• reduced (aerobic) respiration;<br>• fatigue/less energy/not very active; | | 4 |
| | | | | **Total for Question 4 = 13 marks** | |

Type MS · Describe a trend or compare data sets and conclude

Answer 13 of 27 · Question 9 part b.iii · 3 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part b.iii in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 9 | (a) (i) | • reference to platelets;<br>• fibrinogen converted to fibrin;<br>• (fibrin) mesh/crosslinks/fibres formed; | | 3 |
| | (ii) | • reduced blood flow to heart/cardiac muscle / tissue/cells;<br>• less oxygen/glucose delivered;<br>• less (aerobic) respiration/energy released (by cardiac tissue); | | 3 |
| | (b)(i) | size of blood clot/volume of water /aspirin solution used/temperature/shape of blood clot/width of straw; | Allow valid alternatives | 1 |
| | (ii) | use water without aspirin dissolved; | | 1 |
| | (iii) | • the more tablets dissolved the more solution collected;<br>• the more tablets dissolved the quicker the solution is collected;<br>• aspirin dissolves/acts on blood clot;<br>• size of blood clot reduced more quickly with more tablets/blood clot dissolves/breaks down faster with more tablets; | | Max 3 |
| | (c) | Any four from:<br>• aspirin binds to enzyme/aspirin competes (with substrate) for active site;<br>• active site of enzyme changed/blocked;<br>• less/no substrate binds (to enzyme);<br>• reaction involving release of chemicals reduced/stopped;<br>• platelets do not stick together; | | Max 4 |

**Total question 9 = 15 marks**

## Type MS · Describe a trend or compare data sets and conclude

### Answer 13 of 27 · Question 9 part b.iii · 3 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part b.iii in this mark scheme extract

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## Type MS · Describe a trend or compare data sets and conclude

### Answer 14 of 27 · Question 4 part a.ii · 4 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

| number | | | | |
|---|---|---|---|---|
| 4 | (a) (i) | A (bacterium);<br><br>B doesn't cause it<br>C doesn't cause it<br>D doesn't cause it | | 1 |
| | (ii) | • no of cases decreases as vaccination rate increases;<br>• lag in effect;<br>• fluctuations in number of cases;<br>• despite percentage of vaccinations remaining constant; | | 4 |
| | (b) | 2.75 per 100 000;<br><br>$\frac{2.75 \times 3\,450\,000}{100\,000}$;<br>= 95 people; | correct answer without working = 3<br>ECF = 2 | 3 |
| | (c) | • inject weakened/attenuated organism;<br>• causes antibody production;<br>• antibodies remain in blood;<br>• memory cells formed;<br>• on infection antibodies can respond quickly; | | 5 |

Total 13 marks

## Type MS · Describe a trend or compare data sets and conclude

### Answer 14 of 27 · Question 4 part a.ii · 4 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Describe a trend or compare data sets and conclude

### Answer 15 of 27 · Question 3 part c.i · 2 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part c.i in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 3 | (a) (i) | 3 320 000; | | 1 |
| | (ii) | 6.00239;<br>x $10^5$; | Allow 6.0024 | 2 |
| | (iii) | pancreas; | | 1 |
| | (b) (i) | 120 ÷ $1.95^2$/3.8029;<br>31.6; | Allow figure not rounded - 31.5581<br>Allow 31.56/32 | 2 |
| | (ii) | obese; | ecf from previous question | 1 |
| | (c) (i) | • as BMI increases the risk of diabetes increases;<br>• between a BMI of 24-35 the risk of diabetes increases more rapidly/BMI less than 25 the risk increases less rapidly; | Allow any range values between 24-35 | 2 |
| | (ii) | Any two from the following:<br>• balanced diet;<br>• eat less fat;<br>• eat less sugar | Ignore healthy diet<br>Allow reduce alcohol/carbohydrate consumption for mp3 | 2 |

**Total marks 11**

## Type MS · Describe a trend or compare data sets and conclude

### Answer 15 of 27 · Question 3 part c.i · 2 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part c.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Describe a trend or compare data sets and conclude

### Answer 16 of 27 · Question 6 part a.i · 3 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

| Question number | | | | | |
|---|---|---|---|---|---|
| 6 | (a) | (i) | • both increase;<br>• males more than females;<br>• fluctuations;<br><br>accept any two from; | | 3 marks |
| | | (ii) | • layer of actively dividing cells;<br>• can become uncontrolled;<br>• when exposed to ultraviolet/sun light; | | 2 marks |
| | (b) | (i) | 17 - 48;<br>= 31 per 100 000;<br>$\frac{31}{25}$ = 1.2/1.24 per 100 000; | allow 16.7 - 17 and 47.8 - 48 then ECF | 3 marks |
| | | (ii) | 20 per 100 000;<br>$\frac{20 \times 7\,000\,000}{100\,000}$;<br>= 1400; | ECF | 3 marks |

Total 11 marks

## Type MS · Describe a trend or compare data sets and conclude

### Answer 16 of 27 · Question 6 part a.i · 3 marks

2021 NOV · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Describe a trend or compare data sets and conclude

Answer 17 of 27 · Question 3 part b.i · 3 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (i) | arrow from top to bottom; | | 1 |
| (ii) | • A is an artery, thicker muscle wall;<br>• to pump blood from heart/to body;<br>• B is a vein with a valve present;<br>• to control direction of blood flow/to prevent backflow; | If no other marks scored award one for correct identification of both vessels | 4 |
| (iii) | capillary; | | 1 |
| (b) (i) | • decrease in deaths;<br>• in both sexes/males and females;<br>• between 2005 - 2014/over whole range; | | 3 |
| (ii) | • 60 000 000 ÷ 100 000 = 600;<br>• 600 × 640;<br>• = 384 000; | | 3 |

Total 12 marks

## Type MS · Describe a trend or compare data sets and conclude

### Answer 17 of 27 · Question 3 part b.i · 3 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Describe a trend or compare data sets and conclude

Answer 18 of 27 · Question 4 part c.ii · 3 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part c.ii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 4 (a) | | disease causing organism/micro-organism; | | 1 |
| (b) | (i) | any three from<br>• cytoplasm;<br>• cell/plasma membrane;<br>• cell wall;<br>• (loop of) nucleic acid/DNA;<br>• flagellum; | if more than three parts labelled minus one mark for each incorrect label | 3 |
| | (ii) | • distance between A and B = 64 mm/6.4 cm;<br>• 64 × 1000 = 64 000 µm;<br>• magnification = $\frac{64\,000}{2}$ = 32 000; | ecf allow +/- 1 mm | 3 |
| (c) | (i) | • suitable scale, axes labelled with units;<br>• correct way round;<br>• points plotted correctly;<br>• lines joining plotted points;<br>• key to identify lines; | needs to cover half of paper<br>award mp1,2 and 5 if bar chart plotted | 5 |
| | (ii) | any three from<br>• both fall/vaccine becomes less effective over time;<br>• 45-64 years more effective (over all months)/over 65 less effective (over all months);<br>• slowly for first two months;<br>• more/ rapid /steep decrease, between 3-4 months/after month 3 (for over 65); | | 3 |
| | | | | 15 marks |

## Type MS · Describe a trend or compare data sets and conclude

### Answer 18 of 27 · Question 4 part c.ii · 3 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part c.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |

Type MS · Describe a trend or compare data sets and conclude

Answer 19 of 27 · Question 3 part b.i · 2 marks

2022 JUN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (i) | Any five from the following:<br>• place water in the test tube;<br>• take the (starting) temperature of the water;<br>• attach the test tube to a clamp stand;<br>• measure the mass of a piece of food/weigh food;<br>• place food on the mounted needle (under the test tube);<br>• burn the food; (using a Bunsen burner)<br>• take the (final) temperature of the water; | | 5 |
| (ii) | Any two from the following:<br>• volume/mass of water;<br>• distance of the food from the test tube;<br>• mass of food; | ignore amount<br><br>ignore amount/size | 2 |
| (iii) | Any two from the following:<br>• hold the burning food at arm's length;<br>• reference to sharpness of needle;<br>• goggles;<br>• reference to food allergies/don't taste food; | any ref to being burnt/hot water | 2 |
| (b) (i) | Any two from the following:<br>• males need more energy than females;<br>• the amount of energy required decreases with age;<br>• the amount of energy needed by males decreases more (overall) than females; | | 2 |
| (ii) | **D** - Males use more muscle mass than females; | | 1 |
| (iii) | • (2629 – 2103) = 526;<br>• 526 ÷ 2103;<br>• (x 100) = 25 (%); | if divide by 2629 and give 20% then two marks<br><br>Total 15 marks | 3 |

## Type MS · Describe a trend or compare data sets and conclude

### Answer 19 of 27 · Question 3 part b.i · 2 marks

2022 JUN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |

Type MS · Describe a trend or compare data sets and conclude

Answer 20 of 27 · Question 1 part b.i · 2 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 1 | (a) | (i) | molar; | Allow premolar | 1 |
| | | (ii) | C, enamel | | 1 |
| | | (iii) | Any three from:<br>• contains ridges/crevices;<br>• larger surface area/more in contact with food;<br>• food trapped;<br>• provides nutrients for bacteria/reference to bacteria digesting/using food/producing acid;<br>• more difficult to clean/reference to location in mouth (making tooth more difficult to reach); | Reject cusp | Max 3 |
| | (b) | (i) | Any two from:<br>• (both) decrease;<br>• highest % of cases in 1983/lowest % cases in 2013;<br>• (%) 15 year old children with tooth decay always higher/(%) 12 year old children with tooth decay lower;<br>• (%) 15 year old children falls less overall/(%) 12 year old children falls more overall; | | Max 2 |
| | | (ii) | clean/brush/floss teeth (more regularly / properly)/(more/regular) visits to dentist/ reference to fluoride in toothpaste/drinking water/less sugary foods in diet; | Allow better dental hygiene | 1 |

**Total question 1 = 8 marks**

## Type MS · Describe a trend or compare data sets and conclude

### Answer 20 of 27 · Question 1 part b.i · 2 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Describe a trend or compare data sets and conclude

### Answer 21 of 27 · Question 7 part a.ii · 2 marks

2020 JAN · Paper 01 · find part a.ii in this mark scheme extract

| number | | | | |
|---|---|---|---|---|
| 7 | (a) (i) | • 67.5 ÷ 48.3 (1)<br>• 1:1.4 (1) | Full marks for correct final answer | 2 |
| | (ii) | Any two from<br>• more males than females are overweight in all countries (1)<br>• lowest percentage of people between the ages of 18-24 in all countries are overweight (1)<br>• increased in all countries 18 - 64 (1) | | 2 |
| | (iii) | sample size too small (1)<br>not all overweight people identified/sample not representative (1) | | 2 |
| | (b) (i) | Any three from:<br>• pancreas (1)<br>• releases insulin (1)<br>• reference to the liver taking up glucose (1)<br>• glucose converted to glycogen (1) | R sugar<br>R breaks down glucose | 3 |
| | (ii) | Any two from:<br>• high blood glucose increases the concentration of the blood (1)<br>• water moves out of cells (1)<br>• by osmosis (1)<br>• excess water in blood excreted in urine (1) | | 2 |

**Total for Question 7 = 11 marks**

Type MS · Describe a trend or compare data sets and conclude

Answer 22 of 27 · Question 6 part a.i · 2 marks

2020 JAN · Paper 02 · find part a.i in this mark scheme extract

| Question number | | | | | |
|---|---|---|---|---|---|
| 6 | (a) | (i) | • more males than females die (from *S.aureus* and MRSA)(1)<br>• number of deaths from both *S.aureus* and MRSA increase with age(1) | Allow other valid conclusions e.g more deaths from both microorganisms in males/females aged 85 and over | 2 |
| | | (ii) | 416.6 ÷ 1.8(1)<br>231.4:1 (1) | Allow 231:1<br>Allow full marks for correct final answer | 2 |
| | | (iii) | 41.2 + 21.9 = 63.1(1)<br>53 × 63.1 = 3344(1) | Allow 3344.3<br>Allow full marks for correct final answer | 2 |
| | | (iv) | Any two from:<br>• MRSA is resistant (1)<br>• to (many) antibiotics(1)<br>• (MRSA infection) difficult to treat(1)<br>• reference to MRSA infection being (highly) contagious/out of control/easily transferred(1) | | 2 |
| | (b) | (i) | mircroorganisms/bacteria/fungi | Ignore viruses<br>Ignore named antibiotics | 1 |
| | | (ii) | Any six from:<br>• sterilise Petri dishes/agar/nutrient broth(1)<br>• obtain sample(1)<br>• sterile inoculating loop/sterile swab(1)<br>• pass neck of flask containing culture through a flame(1)<br>• smear culture/sample/swab onto agar(1)<br>• reference to adding antibiotics (to agar)(1)<br>• seal Petri dishes/cover with lid(1)<br>• incubate(1)<br>• measure area of clear zone around antibiotic discs(1) | Allow description of how loop/swab is sterilised | 6 |

**Total for Question 6 = 15 marks**

## Type MS · Describe a trend or compare data sets and conclude

### Answer 23 of 27 · Question 2 part a.iii · 3 marks

2023 NOV · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 2 | (a) (i) | • axes labelled with units;<br>• axes correct way round and suitable scale;<br>• points plotted correctly;;<br>• straight lines joined;<br>• lines identified; | time without with<br><br>0 65 110<br>30 120 215<br>60 130 230<br>90 95 200<br>120 80 190<br>150 65 180<br>180 65 170<br><br>don't include in published version | 6 |
| | (ii) | • without = 65 mg per 100cm$^3$;<br>• with = 120 mg per 100cm$^3$; | | 2 |
| | (iii) | any three from<br><br>• without, level starts lower/level lower throughout;<br>• smaller difference between highest and lowest values;<br>• rises to lower levels/rise less quickly;<br>• returns to normal (quicker)/same level 150-180; | ORA for person with diabetes | 3 |
| | (b) | • sample in a test/boiling tube;<br>• add Benedict's reagent/solution;<br>• heat in a water bath;<br>• wear goggles;<br>• brick red colour if glucose present;<br><br>if candidate uses glucose testing sticks<br><br>• sample in a test/boiling tube;<br>• use of testing stick;<br>• dip into urine;<br>• compare colour with chart;<br>• use of goggles; | Accept yellow/green/orange | 5 |

Total 16 marks

## Type MS · Describe a trend or compare data sets and conclude

### Answer 23 of 27 · Question 2 part a.iii · 3 marks

2023 NOV · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Describe a trend or compare data sets and conclude

### Answer 24 of 27 · Question 3 part b.i · 3 marks

2023 NOV · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 3 (a) | (i) | 250:750;<br>1:3; | full marks for correct answer with no working | 2 |
| | (ii) | 17 500 - 5600 = 11900;<br>$\frac{11\,900 \times 100}{5\,600}$;<br>= 210%; | full marks for correct answer with no working<br>213/212.5 = two marks | 3 |
| | (iii) | • other parts of the body/organs;<br>• liver/kidney/lungs/arteries/veins;<br>• needs some blood supplied/blood flows to these organs; | | 3 |
| (b) | (i) | • large/big/huge increase in total flow;<br>• large/big/huge increase in flow skeletal muscles/skin;<br>• decrease for gut; | | 3 |
| | (ii) | • heat generated(body) temperature increases during exercise;<br>• from respiration;<br>• carried by blood/more blood to skin;<br>• (heat) lost through skin/by radiation; | | 4 |

Total 15 marks

## Type MS · Describe a trend or compare data sets and conclude

### Answer 24 of 27 · Question 3 part b.i · 3 marks

2023 NOV · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Describe a trend or compare data sets and conclude

Answer 25 of 27 · Question 3 part e · 2 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part e in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 3 | (a) | pathogens; | | 1 |
| | (b) (i) | 1.2 years; | | 1 |
| | (ii) | • active;<br>• artificial; | | 2 |
| | (iii) | Any two from<br>• weakened/dead/attenuated organisms;<br>• toxins;<br>• antigens;<br>• RNA; | Allow named bacterium/virus<br>Ignore genetic material | 2 |
| | (c) | Any two from<br>• passed from mother;<br>• during breastfeeding/from breast milk;<br>• across placenta; | Ignore umbilical cord | 2 |
| | (d) | Any five from<br>• antibody level falls;<br>• below safe level/to unsafe level;<br>• greater risk of disease;<br>• booster dose increases antibody levels/remains above unsafe level;<br>• person immune/can't catch disease;<br>• increased number of memory cells; | Allow faster response | 5 |
| | (e) | Any two from<br>• antibody level higher;<br>• for longer period/decreases more slowly/remain for many years;<br>• levels of antibody increase faster after booster; | | 2 |

Total 15 marks

## Type MS · Describe a trend or compare data sets and conclude

### Answer 25 of 27 · Question 3 part e · 2 marks

2023 NOV · Paper 02 · extract 2 of 2 · find part e in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |

Type MS · Describe a trend or compare data sets and conclude

Answer 26 of 27 · Question 3 part b · 2 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

| | | | |
|---|---|---|---|
| **3(a)(i)** | B 3.2 - 3.5 kg; | | **(1)** |

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| **3(a)(ii)** | (94 / 210) x 100 ;<br>45%; (2) | Ecf allow correct answer to incorrect figures used in calculation<br><br>Allow 44.76%/ 44.8% for correct final answer | **(2)** |

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| **3(b)** | smoking appears to lower birth mass;<br>more higher birth mass babies born to non-smokers; | ORA for both mp's | **(2)** |

| **Question Number** | Answer | Additional guidance | Mark |
|---|---|---|---|
| **3(c)** | • cigarette smoke contains CO;<br>• combines with haemoglobin instead of oxygen/forms carboxyhaemoglobin;<br>• less oxygen to fetus;<br>• less respiration in fetus;<br>• lower growth rate/cell division;<br>• causes lower birth mass; | | **(5 max)** |

**Total 10 Marks**

Type MS · Describe a trend or compare data sets and conclude

Answer 26 of 27 · Question 3 part b · 2 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

**Paper 2: Subject Human Biology**

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|

Type MS · Describe a trend or compare data sets and conclude

Answer 27 of 27 · Question 2 part d.ii · 2 marks

2020 JAN · Paper 01 · find part d.ii in this mark scheme extract

| number | | | | |
|---|---|---|---|---|
| 2 (a) | (i) | **The only correct answer is A**<br><br>*B is incorrect because it is the cervix*<br><br>*C is incorrect because it is the oviduct*<br><br>*D is incorrect because it is the uterus wall* | | 1 |
| | (ii) | **The only correct answer is D**<br><br>*A is incorrect because it is the vagina*<br><br>*B is incorrect because it is the cervix*<br><br>*C is incorrect because it is the oviduct* | | 1 |
| | (iii) | • oestrogen (1)<br>• lining (1)<br>• ovulation (1) | | 3 |
| (b) | | needed for the growth of bones (1) | | 1 |
| (c) | | ref antibodies (passed to baby)/contains the correct balance of nutrients/reduced risk of diabetes/allergies/obesity later in life/reference to bonding between mother and baby (1) | | 1 |
| (d) | (i) | more breastfed babies in 2010 compared with 1940 (1) | Allow reverse argument<br>A 78% vs 42% difference of 36% | 1 |
| | (ii) | • life expectancy shows a constant increase (1)<br>• despite number of babies breastfed increasing and decreasing (1) | | 2 |

**Total for Question 2 = 10 marks**